

Cyber Security Internship Report

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Company Name: Anonymous India Cyber Defense Academy

Major: Computer Science Engineering (Cybersecurity)

Duration: April 2, 2026 to May 2, 2026

Domain: Ethical Hacking

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Introduction

During my internship at **Anonymous India** from April 2nd, 2026, to May 2nd, 2026, I had the opportunity to work on various ethical hacking projects and tasks. This report outlines the foundational research I conducted, the skills I acquired regarding security frameworks, and the overall learning experience of transitioning from a student to a security trainee.

Projects and Tasks (Level-Beginner)

Assessment 1: Introduction to Cybersecurity & Ethical Hacking

1. Introduction to Ethical Hacking & Methodology

- In the current digital landscape, cybersecurity is no longer optional. During this assessment, I focused on the core principles of Ethical Hacking, which is the practice of bypassing system security to identify potential data breaches and threats in a network.
- Unlike malicious hacking, ethical hacking follows a strict Professional Methodology. I studied the five phases that every penetration tester must follow:
- Reconnaissance: The "info-gathering" phase. This is where I learn about the target's IP addresses and domain info.

- **Scanning:** Using tools like Nmap to find open ports. This is the first time the "attacker" actually touches the target system.
- **Gaining Access:** This is where vulnerabilities are exploited to enter the system.
- **Maintaining Access:** Checking if an attacker can stay in the system (persistent access) even after a reboot.
- **Covering Tracks / Reporting:** In a real-world scenario, this involves clearing logs, but for my internship, the focus is on Reporting—documenting every flaw so it can be fixed.

Outcome: Through this task, I gained a clear architectural understanding of how a professional security audit is structured. I moved from the idea of "randomly hacking" to a systematic approach where I now understand that **Reconnaissance** and **Scanning** are actually the most important phases, as they provide the foundation for a successful and legal penetration test. This has given me the mindset needed to document my findings professionally for a client.

2. Classifications of Hackers (The "Hats")

I learned that hackers are categorized based on their intent and legality. Understanding these "hats" is crucial for defining my own career path:

- **White Hat Hackers:** These are the ethical hackers. We have full permission to hack into systems to find vulnerabilities. Our goal is to protect the organization.
- **Black Hat Hackers:** These are criminals who break into systems for personal gain, data theft, or to cause damage.
- **Grey Hat Hackers:** These individuals might find a bug without permission and then ask the company for a fee to fix it. While they might not have bad intent, their actions are still technically illegal because they lack a contract.

Outcome: This classification helped me recognize that professional cybersecurity is built entirely on **trust and legal authorization**. I now understand that technical skill alone does not make a security expert; it is the adherence to ethical standards and the "Rules of Engagement" that differentiates a professional from a threat actor. This has solidified my goal to pursue a career as a White Hat hacker, operating strictly within legal and corporate frameworks.

3. Legal and Ethical Considerations

One of the most important takeaways from this task is that **skill without permission is a crime**. Professional penetration testing is governed by:

- **The Scope of Work (SoW):** A document that defines exactly what I am allowed to test and what is "off-limits."
- **Non-Disclosure Agreements (NDAs):** Ensuring that any sensitive data I see during a test stays strictly confidential.
- **Compliance:** Following laws like the IT Act (in India) or GDPR to ensure data privacy is respected.

Outcome: This task emphasized that technical expertise must always be secondary to legal compliance. I learned that a successful security professional is defined by their integrity and adherence to the **Scope of Work**. Understanding these legal frameworks has prepared me to handle sensitive data responsibly and ensures that my future testing activities remain fully protected under the law.

4. Common Threat Analysis

I researched the "Big Four" threats that currently dominate the cybersecurity landscape:

- **SQL Injection (SQLi):** I analyzed how an attacker can bypass a login page just by typing 'OR 1=1 --' into a username field. This happens because the backend doesn't "sanitize" the input.
- **Phishing:** I looked at a case study where a simple "Password Reset" email led to a multi-million dollar breach. It proved that the **Human Element** is often the weakest link.
- **XSS & CSRF:** I compared these two; while XSS steals user data (like cookies), CSRF actually tricks the user's browser into performing actions (like transferring money) without them knowing.

Outcome: Researching these "Big Four" threats taught me that security requires both technical logic and human awareness. I learned that most major breaches are preventable through strict input validation and "Defense in Depth" strategies. This analysis provided a practical checklist of vulnerabilities to prioritize during my upcoming lab assessments.

5. Professional Roadmap

For my future goals, I compared three major industry certifications:

- **CEH (v12):** Good for learning the broad range of tools.
- **OSCP (Offensive Security):** This is my long-term goal because it is a 100% practical, hands-on exam.
- **GPEN:** Excellent for learning how to write the professional reports that clients actually pay for.

Outcome: This research provided me with a clear professional roadmap for my future career. I realized that while tools change, the core methodology taught in these certifications remains constant. I have now identified the **OSCP** as my long-term goal, as it aligns with the hands-on, practical skills I am developing during this internship at Anonymous India.

References & Resources Used

1. **Anonymous India Portal:** Roadmap for Assessment 1 Action Items.
2. **OWASP Top 10 (2021/2024):** Reference for SQLi, XSS, and CSRF definitions.
3. **OffSec Training:** Documentation on the Penetration Testing Execution Standard (PTES).
4. **YouTube Reference:** [Ethical Hacking in 12 Hours](#) – Used for visualizing the attack lifecycle.

Personal Learning Note:

While studying the real-world case studies, I was surprised to find that most major hacks aren't caused by "super-tools," but by companies forgetting to update their software or employees using "123456" as a password. It made me realize that basic "Cyber Hygiene" is just as important as advanced hacking

